Sakshi Sharma

AIML & Generative AI Engineer

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Professional Summary

AI/ML and Generative AI Engineer and final-year B.Tech Data Science student (CGPA 9+) with proven experience designing and deploying LLM-powered applications, RAG pipelines, and autonomous agentic systems. Built production-grade AI products including a multi-step agentic job application pipeline (ReAct-style tool-use), a RAG-based legal assistant and an AI carbon credit platform. Proficient in LangChain, FAISS, Hugging Face, prompt engineering, fine-tuning (PEFT, SFT), and agentic workflow design with LangGraph/AutoGen. SIH Finalist, THDCIL intern, and government-funded researcher; seeking to build next-generation AI products powered by foundation models.

Education

B.Tech in Data Science — IPS Academy IES, Indore *2023 - 2027 | CGPA: 9+*
12th Grade — Karnataka Vidya Niketan *2022 – 2023 |*

Work Experience

AI/ML Intern – Binjwa IT Solutions *july 2026-Sept 2026*

- Engineered an intelligent, multi-turn AI assistant for a CRM using Python and LangChain, featuring a robust Role-Based Access Control (RBAC) engine to securely automate and filter employee data queries.
- Working on AIML based project – A unified cost comparision platform for all the LLM, TTS & STT providers.
- Integrated and configured third-party automation workflows, including Canva and Discord, within the Social Media Management (SMM) platform to streamline content creation and communication processes.

AI/ML Backend Engineer (Intern) – THDCIL – Ministry of Power *Sep 2025 – july 2026*

- Engineered an automated document verification pipeline using hash-based tamper detection, reducing manual review effort by ~80%
- Implemented forgery detection models integrated into secure backend workflows.

ML Systems Developer – Autonomous Vehicle for Calamity Rescue (AVCR) – Funded Research Project *Aug 2024 – June 2026*

- Deployed ML inference pipeline on NVIDIA Jetson Nano for real-time survivability assessment in disaster-hit confined spaces
- Achieved 91% classification accuracy using an ensemble of Isolation Forest, LSTM, and Random Forest models on live sensor streams
- Project received institutional funding; selected among top research projects at IPS Academy IES

Projects

Auto Mail Apply — Agentic AI Job Application Pipeline

- Designed and built a fully autonomous multi-step agent that scrapes job listings, ranks candidate-job fit using NLP (cosine similarity + keyword extraction), and auto-composes and sends tailored outreach emails with zero manual input
- Implemented tool-use architecture: agent invokes web scraper, profile matcher, and email composer as sequential tools — a working implementation of ReAct-style agentic workflow. Reduced time-per-application from 20 minutes to under 30 seconds; scales across multiple job boards and candidate profiles

AI-Powered Personal Legal Assistant (RAG + LLM)

- Built a production-grade RAG pipeline combining FAISS vector retrieval with LLM generation for legal document summarization, case strategy advisory, and document classification
- Implemented multi-turn conversational memory and streaming responses for real-time UX

ReelForge — Autonomous AI Short-Form Video Generation Platform

- Architected an end-to-end autonomous video generation pipeline converting natural-language prompts into short-form video reels with LLM-driven scene planning, viral hook generation, and pacing optimization.
- Engineered a multi-modal asset routing and resilient audio engine, dynamically fetching relevant stock media via Pexels and implementing a multi-tier TTS fallback system (ElevenLabs, Edge-TTS, TTS) for synchronized voiceovers.
- Built a distributed headless rendering service using Celery, Redis, and FFmpeg for automated video composition, dynamic caption burning, and audio muxing, streaming real-time stage progress to clients over WebSockets.

Rockfall Prediction & Alert System

- ML classification system predicting rockfall risk in open-pit mines from geological + vibration sensor data; 88% precision on held-out test set
- Real-time alerting pipeline with 10–15 minute early warning ahead of critical risk threshold

Skills

Programming Languages: Python, SQL, React.js
Machine Learning & Deep Learning: Supervised and Unsupervised Learning, Feature Engineering, Scikit-learn, XGBoost, Deep Learning, CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms, Reinforcement Learning, PyTorch, TensorFlow, Scikit-learn, ML flow
Generative AI & LLMs: GPT-4, Claude, Gemini, LLaMA, Hugging Face Transformers, Prompt Engineering, Few-Shot Prompting, Chain-of-Thought Prompting, Instruction Tuning, Fine-Tuning (PEFT, SFT), RLHF (Conceptual), LangSmith, Langfuse, Guardrails, RBAC, Hallucination Mitigation
Agentic AI & RAG: LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), ReAct Agents, Multi-Agent Systems, Tool Calling, Function Calling, Autonomous Workflows, MCP, CrewAI, AutoGen, Conversational Memory, Streaming Responses, STT, TTS, Chatbots, n8n, viasocket
Vector Databases & Search: ChromaDB, FAISS, Embedding Models (sentence-transformers, OpenAI Embeddings), Hybrid Search, Semantic Chunking, Qdrant
Computer Vision & Multimodal AI: OpenCV, YOLOv8, MediaPipe, Image Segmentation, Object Detection, Vision-Language Models
Backend & Deployment: FastAPI, Django, WebSockets, Docker, CI/CD, Model Serving, Redis, Selenium, Authentication (JWT/OAuth), Kubernetes (Basics),
Databases & Analytics: PostgreSQL, MySQL, MongoDB, Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA), Power BI
Cloud & Tools: AWS (EC2, S3), Git, GitHub, Vercel, Google Colab, VS Code
Soft Skills: Problem Solving, Research-to-Product Execution, Rapid Prototyping, Technical Presentations, Hackathon Leadership

Achievements

Winner – UDAAN'26 Phase I and Phase II | Techno Maths Presentation | Science Exhibition 2024 | VyaparX Idea Pitching Competition | Inter-College Chess Competition 2023, 25

Smart India Hackathon (SIH) Finalist | NEC (National Entrepreneurship Challenge) Finalist 2024'25 | Top 20- Clear Skies Innovation Challenge | Top 100 Campus Ambassador, IIT Bombay

Participations hackathons – Bhartiya Antariksha Hackathon, Hacksungum, GDSC Codespire, Pravah 3.0, Hackdata' 25, Build with India, Samsung Solve for Tomorrow, Flipkart Grid 6.0 (Robotics), Snow Hacks

Certifications

- Artificial Intelligence and Machine Learning — NPTEL
- AWS Cloud Operations — Training Badge
- RDBMS PostgreSQL
- Python Tutorial — IIT Bombay

Leadership

- **Founder, E-Cell** — IES IPS Academy: Established the entrepreneurship cell; organized pitching events, mentorship sessions, and cross-college idea competitions